SAINT MICHAEL COLLEGE OF CARAGA

SOURCE ATTRIBUTION OF MICROPLASTICS IN SEAWATER

A Thesis Presented To
The Faculty of
College of Teacher Education

In Partial Requirements for the Degree
Bachelor of Secondary Education Major in Science

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College Of Teacher Education

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2. **List and Explanation**
•Example: *"To my parents, for their love and support, and to my advisor, Dr. [Name], for invaluable guidance."*
3.**Closing (Optional)**
•Example: *"This work would not have been possible without you."*

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Opening Statement
• Express general gratitude for the opportunity to complete the research or thesis.
Acknowledgment of Advisors and Professors
• Thank your thesis advisor, committee members, or professors for their guidance and expertise.

Acknowledgment of Family and Friends
•Thank those who provided emotional or financial support.

Acknowledgment of Institutional or Financial Support
•Recognize scholarships, grants, or organizations that funded or facilitated your research.

Special Thanks (Optional)
Mention individuals who provided significant help, such as peers, mentors, or technical staff.

Closing Statement
•Conclude with a general expression of gratitude.

Example

I am deeply grateful to everyone who contributed to the completion of this thesis. My sincerest thanks to my advisor, Dr. [Name], for your invaluable insights and guidance. To my committee members, thank you for your constructive feedback and support.

To my family, your love and encouragement kept me going through this journey, and to my friends, your support meant the world to me. I would also like to thank [Institution/Organization] for funding this research.

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

Plastics are currently too abundant and simply unstoppable (Castellón-Mena et al., 2024). This makes the Philippines the world's largest source of annual plastic emissions into the ocean (Meijer et al., 2021), which generates 2.7 million tons of plastic waste every year (World Bank Group, 2021). According to S. George (2020), plastics are present in the environment, which means that humans have increasingly relied on them because they are cheap, long-lasting, and lightweight. These factors make plastics essential in life (Pilapitiya & Ratnayake, 2024). However, this growing dependence on plastic has also caused massive environmental (Paterson, 2019). Plastics larger than 5 mm, also known as macroplastics, have become prevalent in aquatic and terrestrial ecosystems (Pacilan & Bacosa, 2022). Since human activities are the primary sources of plastic waste (Retnaningdyah1 & Vidayant, 2025), this pollution leads to various environmental risks. It gets entangled with marine life, chokes mangrove root systems (M. George et al., 2023), and spills toxic compounds and microplastics that enter the seas and food chains (Neelavannan et al., 2025). Identifying the sources and pathways of this degradation is important for creating effective strategies to mitigate marine plastic pollution. Researchers address plastic pollution to support various Sustainable Development Goals, such as health (SDG 3), sustainable cities (SDG 11), consumption patterns (SDG 12), and marine ecosystems (SDG 14), by implementing activities such as waste management, recycling, and encouraging sustainable consumption habits (Muñiz & Rahman, 2025). For the success of a circular economy, plastic waste management is necessary (John N. Hahladakis, 2025). It requires knowledge of its origins and gathering, and the combination of legal, technical, and socioeconomic factors to enhance recycling and recovery. However, despite international measures, considerable amounts of plastic

waste remain mixed with municipal waste, prohibiting recycling and recovery (Todorova et al., 2025).

From the northern to the southern Bulgarian Black Sea coast, plastic pollution is a global issue as plastics enter the sea through various pathways, including rivers, fishing activities, urban runoff, recreational activities, and natural disasters. Once in the ocean, plastics move across borders due to wind, currents, and tides, and often accumulate on beaches, where they can degrade and eventually return to the ocean (Simeonova & Chuturkova, 2020). However, the focus on the spatial distribution of single-use plastics and fishing gear was limited, and there is a need to identify more plastic distribution in the ocean. According to Zambrano-Pinto et al. (2024) claim that human activity is the reason for the plastic pollution found in beach sediments from the Urias coastal lagoon in northwest Mexico. Their results highlighted the size and attributes of plastic pollution in this ecosystem by revealing a prominent presence of macroplastics. The study highlights the importance of better waste management techniques to protect the lagoon's economic and ecological functions. Moreover, the Global South, particularly Southeast Asia, is unequally impacted by the plastic pollution issue.

In Sultan Naga Dimapor, the longest coastline in the province of Lanao del Norte and a high-coastal community, many people raise concerns about the extent of plastic pollution and focus on the amount of plastic that ends up on the seabed. To determine the prevalence, types, and density of macroplastic litter on the coastal seabed, (Pacilan & Bacosa, 2022) collected and analyzed plastic litter from the municipality to establish a baseline for macroplastic litter in the Philippines. In Pujada Bay, Davao Oriental, anthropogenic marine debris and oceanic solid waste significantly affect marine biodiversity and ecosystems, especially in mangrove stands. The researchers in the study by Abreo et al. (2021) collected debris and then compared weights in

different material categories within each area. These researchers used statistical analyses to determine differences in plastics (Abreo et al., 2021; Lumasag et al., 2024). However, neither study provides information on marine debris or plastic sources. Moreover, in Barangay Camagong seawater in Nasipit, Agusan del Norte, bioaccumulation of microplastics in oysters was studied through sample collection and FTIR spectroscopy; however, the sources of microplastics were not included in Lumasag et al. (2024).

In Barangay Camagong, Nasipit, Agusan del Norte, the increasing entry of plastic trash is a pressing ecological and social issue. The improper disposal of domestic and commercial waste has led to continuous plastic pollution in coastal waters, affecting aquatic organisms through plastic degradation (Kibria et al., 2023). From the study of Lumasag et al. (2024), a residential site in Barangay Camagong, Nasipit, Agusan del Norte, indicated the presence of microplastics, specifically D polyvinyl stearate and PVC with Citric Ester in oyster flesh. This finding underscores the urgent need to identify and quantify the sources of microplastic contamination in the seawater. Working on a single site, such as Barangay Camagong, offers benefits for experimental research. Studies have shown that sites with human waste are often sources of aquatic pollution (Kye et al., 2023). Plastics enter the ocean through various pathways, including land-based sources from domestic waste and sea-based sources from fishing activities and maritime shipping (Thushari & Senevirathna, 2020). Mapping and classifying microplastic sources, especially in Barangay Camagong will provide significant information on the sources of microplastics and the migration of macroplastics within the barangay's marine ecosystem (Pierard et al., 2022).

The findings of this research will be advantageous to the stakeholders living in Barangay Camagong, Nasipit, Agusan Del Norte, providing evidence on macroplastics and their dispersal of microplastics, which contribute to the bioaccumulation of microplastics in oysters. The results of

this study will benefit the stakeholders residing in Barangay Camagong, Nasipit, Agusan Del Norte, by providing evidence of macroplastics and their dispersal of microplastics, which is essential to understand the source attribution of seawater microplastics and to determine the origin of the pollutants and their transport in the marine environment (Ashrafy et al., 2023; Pierard et al., 2022). The Municipal Environment and Natural Resources Office (MENRO) and the Local Government Unit of Nasipit can use the findings to enhance coastal cleanup drives and improve proper community waste disposal, thereby reducing environmental dumping throughout the municipality. Saint Michael College of Caraga hopes to apply the results of this research to strengthen and innovate its current waste management programs, encouraging sustainable practices within the institution and communities. The results of this study are to provide detailed, evidence-based information that will help residents, policymakers, and other stakeholders develop more effective solid waste management strategies, thereby supporting better environmental quality and public health gains.

1.2 Review of Related Literature

This chapter will provide the foundational background for the research, incorporating ideas and support from various scholarly works. The literature review aims to study macroplastic pollution in seawater in Barangay Camagong, Nasipit, Agusan del Norte and address the significant reasons and factors that contribute to plastic accumulation, the description of macroplastic types, sizes, sources, and polymer composition, contamination of coastal waters by land-based and ocean-based plastics, the possible environmental and ecological risks of macroplastic pollution, and the monitoring and analytical methods to assess plastic pollution. The findings will inform key stakeholders, including the Local Government Unit of Nasipit, the Municipal Environment and Natural Resources Office of Nasipit, and the Barangay Captain of Barangay Camagong, Nasipit,

Agusan del Norte. The researchers will present the findings to establish connections between various theories and research outcomes.

Literature of the Study

Macroplastic

Macroplastics are becoming an environmental concern due to their presence in ecosystems and the damage they cause to marine, freshwater, and even terrestrial environments (Al-Zawaidah et al., 2021). Macroplastics have many sources, including daily single-use products like bags, bottles, cups, and straws, as well as fishing litter such as nets and lines (Bimali Koongolla et al., 2018). Single-use plastics, especially, are a significant issue; they pollute water heavily, often because they are thrown around recklessly or come from domestic waste (Blettler & Mitchell, 2021).

Over time, environmental factors like UV light, mechanical stress, and microbial breakdown break these larger plastics down into smaller pieces, ultimately forming microplastics —particles smaller than 5 mm. This process of transformation has been well established in research on the degradation of plastics in aquatic ecosystems (Alimba & Faggio, 2019). The breakdown of macroplastics into microplastics increases pollution and complicates mitigation efforts, as smaller particles are more challenging to detect and remove from ecosystems.

Categories of Macroplastics

Plastic waste is an issue in the world's oceans and waterways (The National Research of the Philippines, n.d.). Individuals use plastic items primarily for basic needs, such as food security, transportation, housing, and sanitation (Uknowledge & Meyer, 2021). Poor waste management and improper disposal of plastic materials result in their accumulation as waste, much of which ends up in the ocean, causing pollution (Abalansa et al., 2020). Plastic waste categories that find

their way into the sea are (a) food packaging, (b) disposable utensils, (c) food containers (d) cloth, (e) plastic caps, (f) napkin and diapers, (g) ropes and nylon, (h) cigarette butt, (i) plastic fragments, (j) sando bags, (k) styrofoam, (l) medicinal packaging, (m) sack, (n) plastic bottle, (o) cellophane and (p) other plastics (Pacilan & Bacosa, 2022; Sajorne et al., 2021). In the Adriatic Sea region, degrades or fragments of plastics happen over time, resulting in an abundance of microplastic debris (Gomiero et al., 2019). The sampling segregation was based on what types of plastics occur in terms of size, polymer type, and shape (Gomiero et al., 2019; Jemaa et al., 2021; Maharana et al., 2020). A study by Râpă et al. (2024) revealed that, among the marine plastic litter collected, single-use plastics (SUPs), such as bottles, bags, food wrappers, and packaging materials, were the most frequent items. Also, under the Atlantic Ocean, broken plastics, such as intact bottles, bags, and nets, are the dominant type of plastic waste. In the mangrove areas around Cebu Island, Philippines, fragments like plastic bags and packaging items are the most common types of plastic litter originating from nearby communities and fishing activities (Paler et al., 2022). The National Oceanic and Atmospheric Administration concluded that the commonly found plastics in the ocean include cigarette butts, food wrappers, beverage bottles, straws, cups and plates, bottle caps, and single-use bags.

Microplastics

Given that a significant proportion of packaging is designed for single use, it frequently accumulates in the environment (Tramoy et al., 2025). The degradation of these larger plastic items results in smaller particles, commonly referred to as microplastics. Research conducted in Laguna de Bay indicates a positive correlation between microplastic concentration and population density. In this ecosystem, the majority of microplastics are fibrous in nature (57%), with a predominance of blue coloration (53%). The most prevalent polymers identified include

polypropylene (PP), ethylene vinyl acetate copolymer (EVA), and polyethylene terephthalate (PET), collectively accounting for 65% of the total microplastic composition (Arcadio et al., 2022). Furthermore, a study performed in Binh Dinh, Vietnam, analyzed microplastic content in the digestive systems of various fish species, categorizing the particles by type, size, and color. Other marine organisms, specifically mollusks like oysters, are identified as indicators of microplastic pollution due to their filter-feeding habits (Villaceran et al., n.d.; Wootton et al., 2022). Microplastics were detected in all specimens, with microfibers being the most abundant type, and polyethylene, polyvinyl ether, and polymethacrylate identified as primary constituents. A research gap remains in the comprehensive understanding of the long-term ecological and human health implications associated with this microplastic contamination (Tran et al., 2023). While microplastics are recognized as complex, multifaceted pollutants, the precise source of specific plastics of their accumulation within the tissues and organs of marine organisms requires further qualitative and quantitative investigation.

Polymer Types of Plastics

According to the research conducted by Pacilan & Bacosa (2022) and Sajorne et al. (2021), plastic polymer categories are polyethylene terephthalate (PET), high-density polyethylene (HDPE), polyvinyl chloride (PVC), low-density polyethylene (LDPE), polypropylene (PP), polystyrene (PS), polyamide (PA), multilayer, and others. The results of the study indicate that knowing the types of polymers traces the origins of pollution and identifies the sources of the plastics, whether from residential areas, mangroves, beaches, or non-residential locations in the beaches of Puerto Princesa, Palawan Island, and in the coastal sea beds of Sultan Naga Dimaporo, Lanao Del Norte, Philippines. In Myanmar's Ayeyarwady coastal regions, the research also found that the two most prevalent plastic polymers in marine litter were polypropylene (PP) and

polyethylene (PE). Such materials are associated with consumer products such as food and beverage packaging (LYNN et al., 2025). Different polymers also have distinct properties that affect buoyancy, rates of degradation, and interactions with seas. For instance, polyethylene (PE) and polypropylene (PP) float and may drift long distances, whereas polyvinyl chloride (PVC) sinks and settles in sediments (Pakhomova et al., 2020). It is crucial to determine polymer types from plastic because various polymers have varied densities and degradation rates. This determines whether they float, sink, or disintegrate, hence having an influence on their distribution in marine environments (Dumon, 2025).

Coastal Population Density

Coastal population density understands the pressures placed on the marine environment (Reimann et al., 2023). Most beaches in the Philippines were classified as filthy to extremely bad by the Clean-Coast Index (CCI) in several studies on plastic pollution monitoring (Sajorne et al., 2021). Many studies have identified a positive relationship between human population density and macroplastic abundance. Areas with higher coastal populations generate more plastic waste, which enters the marine environment through poor waste management, sewage, and surface runoff (Peng et al., 2020). Recent regional assessments confirm that densely populated shorelines consistently report higher macroplastic loads than sparsely inhabited areas (Acot et al., 2022).

Higher coastal population density increases the likelihood that mismanaged waste will enter rivers and coastal waters, thereby leading to greater macroplastic accumulation along shorelines (Pinto et al., 2024). Over time, these macroplastic fragments, caused by solar radiation, mechanical abrasion, and wave action, produce secondary microplastics that persist in marine ecosystems (Thushari & Senevirathna, 2020). Thus, in densely populated coastal areas such as Barangay Camagong, Nasipit, Agusan del Norte, poor waste management, coupled with increasing

human activity, can significantly elevate macroplastic deposition, which may later fragment into microplastics, affecting marine life (Marmara et al., 2023).

While substantial progress has been made in quantifying coastal population density and macroplastic pollution, gaps remain in integrating these variables. Many studies focus on demographic analysis or pollution monitoring, but few explicitly link standardized population measures to plastic abundance in a comparative framework (Liu et al., 2022).

Environmental Factors

Large plastic debris, such as macroplastics, can cause entanglement, ingestion, or blockage of digestive tracts in marine animals, including fish, sea turtles, seabirds, and marine mammals (Marmara et al., 2023). Macroplastics act not only as direct hazards but also as pioneers of microplastics, thereby indirectly increasing the burden of plastic pollution in marine ecosystems (Bacha et al., 2023). Even if ingestion does not lead to immediate death, it can produce sublethal, chronic effects on the organisms (Lechthaler et al., 2020).

Marine animals are known to mistake floating macroplastics for prey or become entangled in larger plastic items. The presence of macroplastics contributed by rivers adds to the volume of items in the marine environment that can entangle or be ingested by fish, marine mammals, seabirds, turtles, etc. (Al-Zawaidah et al., 2021). Macroplastics remain the primary contributors to plastic pollution, both as physical pollutants and as the source materials that produce microplastics. The environmental consequences of macroplastic pollution in marine environments are wide-ranging, affecting individual organisms, habitats, and ecosystem functions (Inocente et al., 2025). These macroplastics act as mobile pollutants, entering rivers from earthly sources and distributed along river channels, banks, and floodplains (Dalu et al., 2025). These

interactions near water bodies indicate that macroplastics are redistributed along river corridors, becoming hazards to wildlife in those zones (Blettler & Mitchell, 2021).

According to Simeonova & Chuturkova (2020), household plastics, including food wrappers, bottles, and bags, are major contributors to terrestrial and marine pollution. Improper disposal, such as littering and inefficient waste management, exacerbates this issue, allowing plastics to enter rivers, beaches, and oceans. Industrial activities also contribute to macroplastic pollution in the marine environment and coastal areas, particularly through plastic packaging and disposable materials used in manufacturing and distribution processes, including packaging from industries like food production, medicine, and chemicals, which have a link to plastic debris found in marine environments (Kumar et al., 2021).

Seawater

Seawater accounts for more than 70% of the Earth's surface area and is crucial in providing sustenance for marine life as well as human activities (PLASTIC POLICIES IN THE PHILIPPINES Country Profile 2, 2025). As a result, fisheries, transportation, recreation, and industry are closely linked to economic and social development (Koelmans et al., 2017). However, in recent years, pollution from sources like factories, sewage, and plastic waste has threatened seawater quality (Li et al., 2023). The pollutants consequently lead to the accumulation of plastics and toxic substances, harming marine food webs and lowering water quality (Muñiz & Rahman, 2025). For instance, massive plastic objects in the ocean can destroy habitats, trap marine species, and distribute invasive species (Galgani et al., 2021). Moreover, contaminants on plastic items can enter the food web when humans consume seafood, potentially affecting human health (de Carvalho-Souza et al., 2018). Thus, to safeguard marine environments and protect coastal

communities from hazards, it is imperative to manage and monitor seawater quality (Dalu et al., 2025).

Seabed

According to Zhu et al. (2024), researchers introduced a global estimate of macroplastic pollution on the ocean floor, highlighting it as a significant source, accounting for 3-11 million metric tons. Macroplastic accumulation is also concentrated in continental margins and high-density areas, where low degradation rates and high impacts on marine life are the norm. Based on the research by Morales Jiménez et al. (2023). Macroplastic accumulation on reef substrates depends on hydrodynamic conditions, object shape, polymer type, and the extent of dirt. Onshore pollution sources, such as industrial discharges, plastic waste, and human waste, particularly affect the ocean bottom ecosystem, highlighting the importance of enhanced marine waste management procedures and increased environmental consciousness (Putera et al., 2024).

Industrial Discharge

Industrial discharges contribute heavily to elevated levels of macroplastics in seawater. Most coastal businesses discharge wastewater comprising plastic residues, packaging, and synthetic fibers into the sea without treatment (Koelmans et al., 2017). Regions with high numbers of factories and poor waste management experience particularly significant amounts of macroplastics from manufacturing, fisheries, and shipping (Requiron & Bacosa, 2022). Such pollution harms marine life, lowers water quality, reduces fish populations, and harms tourism (Zhang et al., 2023). Weak regulations and poor enforcement worsen the problem, underscoring the need for improved production practices and effective wastewater treatment to reduce macroplastic pollution in the ocean (Dalu et al., 2025).

Health Risk

Macroplastics remain present in seawater, with increasing health impacts on marine life and humans (Koelmans et al., 2017). During decomposition, large plastic waste produces smaller pieces that can absorb toxic chemicals, including heavy metals, persistent organic pollutants (POPs), and pathogens, from water (Lakhiar et al., 2024). When sea creatures consume these plastics, toxins are absorbed into their systems and can enter the human food chain through seafood (de Carvalho-Souza et al., 2018). Chronic exposure to such pollutants may result in endocrine disruption, leading to abnormal menstrual cycles, infertility, weight gain, mood changes, premature puberty, and hormonal imbalances, such as decreased testosterone or estrogen (Lam et al., 2020). Liver injury may present with abdominal pain or swelling, icterus, vomiting, dark-colored urine, tiredness, and loss of appetite due to the buildup of harmful compounds, including heavy metals and persistent organic pollutants (Sobhani et al., 2022). Neurological conditions resulting from microplastic exposure include headaches, memory loss, sleep disturbance, dizziness, and mood or behavioral changes due to toxins that affect brain activity (Prüst et al., 2020 & Rahman et al., 2023). In the meantime, gastrointestinal problems often arise when organisms consume microplastics, leading to inflammation or interference with the gastrointestinal tract, causing stomach pain, bloating, nausea, constipation, and diarrhea, and altering the gut microbiota (Jiang, 2018 & Stephanie L Wright, 2017). All these facts prove that there is an urgent need to enhance pollution control and raise people's awareness to decrease human exposure to plastic toxins (Winton et al., 2020).

Literature of the Method

Sample Collection Timing

The recommended days for collecting macroplastic samples are spread over three days, with sampling conducted on three weekends to capture variations in plastic deposition influenced by human activities and tidal changes (Pacilan & Bacosa, 2022). The weeks' duration is also important to determine the clean coast index in a specific site within the study area (Acot et al., 2022, Pacilan & Bacosa, 2022).

To minimize contamination during sample collection, researchers collecting plastics from seawater are advised to wear Personal Protective Equipment (PPE), including gloves, boots, and masks, to prevent personal exposure and sample contamination (Kannan et al., 2023). Each sample should be placed in clean, labeled polyethylene containers immediately after collection. The researchers must carefully avoid disturbing sediments or marine organisms during debris retrieval (S Lippiatt, 2020). Additionally, collecting macroplastics, researchers walked along transect lines during low tide to collect plastics from the shoreline and shallow areas (Acot et al., 2022, Pacilan & Bacosa, 2022, Sajorne et al., 2021). For deeper or less accessible areas, boat transects were conducted during high tide to ensure all parts of the area were covered (Dalu et al., 2025).

Line Transect

This study will use the line transect at low tide as one of its independent variables (Acot et al., 2022, Pacilan & Bacosa, 2022, Sajorne et al., 2021). According to Ryan (2022), this approach entails orderly gathering visible macroplastic waste along a predetermined coastal line during low tide. When sampling at low tide, a larger intertidal region is exposed, facilitating the identification and recovery of stranded plastics larger than 5 mm (Karthik et al., 2021). Typically,

each transect is 100 meters long and 5 to 10 meters wide on either side, guaranteeing uniform sampling coverage and site comparability (Acot et al., 2022; Pacilan & Bacosa, 2022). The line transect approach has been widely used in coastal monitoring programs to ensure consistent and repeatable data collection (Gaboy et al., 2022; Galgani et al., 2021).

Boat Transect

This study will use the boat transect at high tide as an independent variable (Dalu et al., 2025). With this approach, a boat travels pre-planned transect routes to gather floating or submerged macroplastic waste from deeper or less accessible coastal locations. By sampling at high tide, researchers can access offshore areas that are hidden at low tide, ensuring thorough coverage of the marine ecosystem. According to the Mekong River Commission (2023), each boat transect usually travels a straight line parallel to the beach, with standardized separations between sample locations to ensure uniformity. According to this method, the use of a net is crucial to capture the floating macroplastic waste, which is 0.1 mm - 0.3 mm (Mekong River Commission, 2023). This transect reduces bias and improves the representativeness of the data gathered under different tidal circumstances. Boat-assisted transects are very useful when evaluating the quantity, makeup, and distribution of macroplastics in open and semi-enclosed water environments (Galgani et al., 2021).

Sample Categorization

According to the Department of Environment and Natural Resources (DENR), samples must be stored in clean, labeled polyethylene bags or stainless containers right after collection. (PLASTIC POLICIES IN THE PHILIPPINES Country Profile 2, 2025). Before packaging, plastics should be kept at room temperature for four hours with no air movement. This is necessary to prevent collected plastics from contaminating one another and to dry out any moisture (Villegas-Camacho

et al., 2024). Once dried, samples are sorted and categorized to help analyze their type, size, source, and degradation state (Galgani et al., 2021). The collected macroplastics will be weighed per plastic category, ranging from 0.11 g - 4.70 g or from 0.068 mg - 0.459 mg\cm2 (Gerritse et al., 2020), depending on the density of each plastic. The size of the plastic sample will be carefully cut, no larger than 5 mm, using a Vernier caliper to maintain the same size in each cut sample (Villegas-Camacho et al., 2024). The plastics then classified based on a modified framework from the study by Sajorne et al. (2021) covering everyday items such as (a) food packaging, (b) disposable utensils, (c) food containers, (d) cloth, (e) plastic caps, (f) napkins and diapers, (g) ropes and nylon, (h) cigarette butts, (i) plastic fragments, (j) sando bags, (k) styrofoam, (l) medicinal packaging, (m) sacks, (n) plastic bottles, (o) cellophane, and (p) other plastics. Each item will be cleaned of remaining sediments or organic material using Hyclin Plus, followed by an initial rinse with tap water (Villegas-Camacho et al., 2024). The researchers then used Milli-Q-grade distilled water to avoid interference during classification and FTIR analysis (Constant et al., n.d.; Villegas-Camacho et al., 2024).

Sample Transportation

Immediate and careful handling is essential to maintain the integrity of collected macroplastic samples. Proper transportation of macroplastic samples is necessary to prevent contamination, photodegradation, and moisture-related changes (Wang et al., 2024). Samples will be placed in clean, labeled polyethylene bags or containers immediately after collection (Koelmans et al., 2017). During transportation, samples will be sealed and protected from ultraviolet (UV) light, heat, rain, and physical contact to maintain their physical and chemical integrity (Koelmans et al., 2017).

Polymer Analysis Using FTIR Spectroscopy

At MSU-IIT Iligan, Fourier Transform Infrared (FTIR) spectroscopy will be used to identify the polymer composition of macroplastic samples (Arcadio et al., 2022, Villegas-Camacho et al., 2024). This technique produces an infrared spectrum for each sample. Also, the technique has identification aided by comparison with reference spectra (Kroon et al., 2018). Before analysis, researchers will clean and air-dry the samples to remove any debris or moisture that could change spectral results (Dalu et al., 2025 & Nayebi et al., 2023). FTIR analysis identified polymers such as polyethylene (PE), polypropylene (PP), polystyrene (PS), and polyethylene terephthalate (PET).

1.3 Objectives of the Study

The main goal of this research is to analyze the macroplastic litter, particularly the source attribution of these plastics in the seawater of Barangay Camagong, Nasipit, Agusan del Norte. The particular objectives of the study are as follows:

1. to determine the most common plastic per site in the study area;
2. to identify and categorize macroplastics within the study area;
3. to determine the sources of polymer type of the collected macroplastic waste;
4. to relate the polymer type of the macroplastic sample to the microplastic presence in specific seafood; and
5. to inform stakeholders with feedback on the study's findings.

1.4 Scope and Limitations

This study will focus on identifying the sources of microplastics in seawater. The researchers will conduct the study in the coastal waters of Barangay Camagong, Nasipit, Agusan del Norte, selected for their proximity to potential microplastic sources. The categories of plastics that the researchers will collect are only those greater than 5mm in size, visibly seen along

mangrove, residential, and non-residential sites (Pacilan & Bacosa, 2022). The collected plastics will be categorized for polymer test analysis based on the three most prevalent items that will be found in the study area.

The plastic samples will be delivered to Mindanao State University, Iligan City, for polymer testing using Fourier Transform Infrared Spectroscopy (FTIR) micro-spectroscopy analysis. This technique confirmed the identification and classification of microplastics in oysters, as reported by Lumasag et al. (2024). However, this technique has limitations for identifying every particle, particularly those that are very small or have compositions similar to those of naturally occurring substances.

1.5 Definition of Terms

To further understand the contents of this research, the following are the keywords indicated in this study:

Attribution. Refers to tracing the source of microplastics in Barangay Camagong, Nasipit, Agusan del Norte in seawater.

Boat transect. Refers to the method of collecting plastics in a body of water during high tide.

FTIR. Stands for Fourier Transform Infrared spectroscopy, which can identify the type of polymer in plastic samples.

Line transect. Refers to the method of monitoring macroplastic debris along a long, straight shoreline in Barangay Camagong, Nasipit, Agusan del Norte.

Mangrove site. This sampling site refers to the vast mangrove forest near and on the shoreline of Barangay Camagong, Nasipit, Agusan del Norte.

Macroplastic. Refers to plastic items larger than 5 millimeters (0.5 cm) and visible during high and low tides, which the researchers will collect from seawater in Barangay Camagong, Nasipit, Agusan del Norte.

Microplastic. Refers to tiny plastic particles, typically less than 5 millimeters in size, that accumulate in marine animals.

Non-residential site. This sampling site refers to the absence of any housing in Barangay Camagong, Nasipit, Agusan del Norte, in the seawater.

Polymer Type. Refers to the classification of the sources, structures, and properties of the materials found in Barangay Camagong, Nasipit, Agusan del Norte.

Residential site. This sampling site refers to the community living near the coastal area in Barangay Camagong, Nasipit, Agusan del Norte, in the seawater.

Seabed. Refers to the ocean floor, where plastic debris settles and becomes visible during low tide.

Seawater. Refers to the water from Barangay Camagong, Nasipit, Agusan del Norte, which the researchers will use for macroplastic collection.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter discusses the methodology to be used in this study, including the experimental research design, research materials and methods, research environment, population, sample, and ethical considerations. This study will focus on collecting macroplastics from Barangay Camagong seawater to confirm the plastic polymer type that contributes to the bioaccumulation of microplastics in oysters (Villaceran et al., n.d.; Wootton et al., 2022) and in the same seawater of Barangay Camagong, as observed from the study of Lumasag et al. (2024). By following procedures and ethical practices, the research will ensure accurate results and provide insights into the sources of microplastics.

2.1 Research design

This study is an experimental research that aims to identify the sources of microplastics by analyzing the polymer types of macroplastics present in seawater. Specifically, researchers will collect and categorize plastic materials from three selected sampling sites in the seawater of Barangay Camagong, Nasipit, Agusan del Norte. The collection will be done for three weekends, the first week for collecting plastic samples and the remaining weeks for the survey and observation process of plastic accumulation in the study area (Gontarek-Castro et al., 2026). To further analyze the composition of these plastic samples, with the three most prevalent plastic items within the study area, Fourier Transform Infrared (FTIR) Spectroscopy will be used to determine their polymer types. This analysis will happen at the Centers for Sustainable Polymers (CSP) Laboratory at Mindanao State University, Iligan Institute of Technology (MSU-IIT). By comparing the polymer types found in macroplastic samples to those commonly identified in microplastic pollution, the study aims to establish a link between macroplastic waste and the

presence of microplastics in seawater organisms. Through this method, the research aims to attribute microplastic contamination to specific sources, providing a scientific basis for targeted interventions in marine pollution management.

The researchers will complete the method to secure approval from the appropriate authorities, including the Municipal Environment and Natural Resources (MENRO), the Local Government Unit of Nasipit, and the Barangay officials of Camagong, Nasipit, Agusan del Norte. The researchers will collaborate to ensure proper plastic collection at each site. Being sure about the collection involves determining the specific area size from the transect. The researchers will also seek approval from MSU-IIT for polymer testing of the samples to be collected. The researchers will submit the permission requests before the sample collection date. During sample collection, researchers will minimize environmental impact by wearing PPE to avoid skin contamination from the plastics and seawater (CDC Logo The National Institute for Occupational Safety and Health (NIOSH), 2022). The researchers will send the plastic samples to MSU-IIT for polymer testing, where established protocols guide them to ensure accurate and reliable results.

Materials

The materials needed for the experiment are as follows:

Disposable gloves. The researchers will use this to collect plastic samples from the three sites in Barangay Camagong sea to avoid contamination.



Figure 1. Disposable gloves

Face masks. The researchers will use this during plastic collection to ensure safety and prevent inhalation of microplastic particles.



Figure 2. Face masks

Trash bags. The researchers will use this to collect and properly categorize plastics from the seawater in Barangay Camagong.



Figure 3. Trash bags

Trash bin. The researchers will use this to store and contain the collected plastics during the collection process.



Figure 4. Trash bin

Stainless steel ruler. The researchers will use this to measure the length of macroplastics accurately.



Figure 5. Stainless steel ruler

Analytical balance. The researchers will use this to measure the weight of the macroplastics being collected.



Figure 6. Weighing scale

Tongs. The researchers will use this to sort plastics after manual collection.



Figure 7. Tong

Fourier Transform Infrared Spectroscopy (FTIR). It is an analytical technique used to identify and verify the type of plastic in plastic samples.

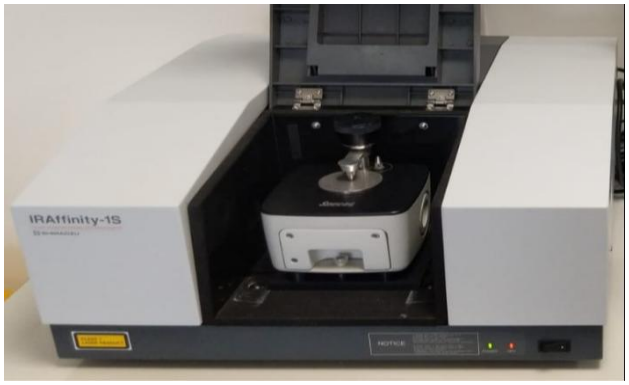


Figure 8. Fourier Transform Infrared Spectroscopy

Steel tape. The researchers will use this as a tool to measure the line transect during plastic sample collection.



Figure 9. Steel tape

Waterproof boots. The researchers will use this to protect feet from sharp objects, mud, and debris in seawater during plastic collection in the low tide season.



Figure 10. Waterproof boots

Wooden stakes. The researchers will use this to mark the start and end of each transect.



Figure 11. Wooden stakes

Global Positioning System or GPs. The researchers will use this to record the transect position, allowing for the repetition of sampling during subsequent collection sessions.



Figure 12. Global Positioning System

Boat. The researchers will use this only during high tide to access floating plastics in Barangay Camagong's seawater.



Figure 13. Boat

Oar. The researchers will use this to manually navigate the boat during the collecting operation in Barangay Camagong's seawater.



Figure 14. Oar

Fishing net. The researchers will use this to collect floating plastics and debris found on the water surface during the plastic collection operation.



Figure 15. Fishing net

Stainless steel container. The researchers will use this to store dried plastic to prevent contamination.



Figure 16. Stainless steel container

Polyethylene storage container. The researchers will use this container to store dry-collected plastics to avoid prolonged exposure to the environment.



Figure 17. Polyethylene storage containers

Vernier Caliper. The researchers will use this to measure plastic in its appropriate size for polymer testing.



Figure 18. Vernier Caliper

Milli-Q Distilled Water. The researchers will use this to rinse plastics and remove surface contaminants, salts, and residues.



Figure 19. Milli-Q Distilled Water

Hyclin Plus. The researchers will use this as a disinfectant and cleaning solution to sanitize and remove organic contaminants, biofilms, or microbial residues from the plastic surface.



Figure 20. Hyclin Plus

Scissor. The researchers will use this to measure plastic in its appropriate size for polymer testing.



Figure 21. Scissor

Disposable Surgical Gown. The researchers will use this to minimize the harm and to prevent contamination, both to the plastics being collected and to the person collecting plastic.



Figure 22. Disposable Surgical Gown

Safety Goggle. The researchers will use this to protect the eyes from contact with contaminated water and physical debris.



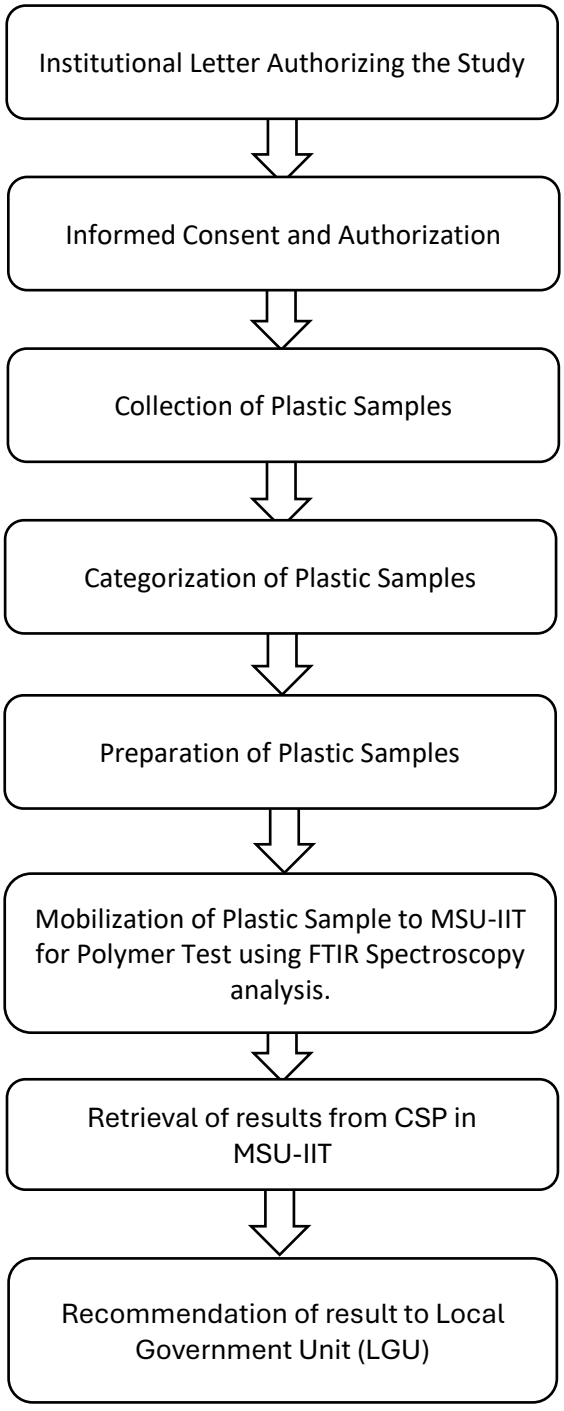
Figure 23. Safety Goggle

Face Shield. The researchers will use this as a physical barrier to help protect the face and associated mucous membranes.



Figure 24. Face Shield

Process flow



Methods

The following procedures outline the collection of plastic samples from seawater at three designated sites in Barangay Camagong, Nasipit, Agusan del Norte.

Collection of Plastics

1. On the collection day, the researchers will gather at District 1, Barangay Camagong, Nasipit, Agusan del Norte to disseminate materials and PPE for collecting plastics during low tide, such as a trash bin, a trash bag, gloves, masks, surgical gowns, goggle, face shields, and boots, and during high tide, researchers will rent a boat with an oar and use fishing net for collecting plastics on the surface of the seawater.

2. The researchers will be collecting plastics in residential, non-residential, and mangrove sites for three weekends.

3. The researchers will wear Personal Protective Equipment (PPE) to collect plastics in the Barangay Camagong plastic samples.

4. The researchers will weigh the collected plastics.

5. The researchers will rinse the collected plastics with Hyclin plus, tap water, and Milli-Q distilled water, and then pat dry using a clean cloth and keep the plastics at room temperature without air currents.

6. Then, the dried plastics will be weighed.

7. Other non-plastic debris that the researchers will collect will be sealed in a trash bag and will be disposed of properly.

Categorization of Plastics

1. After the drying and weighing process, the researchers will sort the collected plastics using tongs to determine which plastic category they belong to, and which among the plastic items are the most common in the study area.

2. The researchers will also weigh the plastics into sample.

Preparation of Plastics

1. The researchers will carefully cut the plastics into pieces that are not more than 5 mm in size to replicate the exact properties of microplastics.

2. Plastic samples will be stored in a clean stainless-steel container and placed in the polyethylene storage container.

3. Plastic samples will then be stored in a dust-free environment not exceeding room temperature, ready for transportation to The Center for Sustainable Polymers (CSP) laboratory in Bldg. 23, Mindanao State University - Iligan Institute of Technology, Andres Bonifacio Avenue, Tibanga, 9200 Iligan City, and used FT-IR, or Fourier Transform-Infrared Spectroscopy, for polymer test analysis, which will take approximately 5-6 hours.

2.3 Research environment, population, and sample

The study area will be in the seawater of Barangay Camagong, Nasipit, Agusan del Norte. The study site will be located in a coastal area inhabited by residents and mangroves, around 8°58' North latitude and 125°20' East longitude, in the northern region of Mindanao, within the province of Agusan del Norte. The researchers will set up one station with three sampling sites in District 1, Barangay Camagong. Their locations will be determined using Google Maps. Site 1 (8°58'27.6"N 125°20'14" E) will be the residential site of the NARWHAL and the coastal community. The collection of plastic at this site will be during low tide. On the other hand, during high tide,

collection of plastic is at Site 2 (8°58'27.9"N 125°20'14.2" E), where it is the mangrove site and most of the fisherfolk resided, and Site 3 (8°58'30.7"N, 125°20'09.2"E), which is the non-residential site, but near the Nasipit Port, where the active industrial shipping is located. According to the study by Lumasag et al. (2024), they selected these sites in Barangay Camagong because of potential environmental sources of microplastic pollution, particularly the evidence of macroplastics from human activities, such as macroplastic discharges from the port due to commercial activities and residential communities, which contribute to the presence of microparticles in the seawater.

For this study, the population comprises all macroplastic debris in the seawater under investigation that is larger than 5 millimeters and capable of undergoing physical, chemical, and biological degradation to form microplastic particles less than 5 millimeters (Yousafzai et al., 2025). These plastics serve as the secondary sources of macroplastics that the researchers will collect and sort into categories, which the researchers modified from the study of Sajorne et al. (2021), which include everyday items such as (a) food packaging, (b) disposable utensils (c) food containers, (d) cloth, (e) plastic caps, (f) napkin and diapers, (g) ropes and nylon, (h) cigarette butt, (i) plastic fragments, (j) sando bags, (k) styrofoam, (l) medicinal packaging, (m) sack, (n) plastic bottle, (o) cellophane and (p) other plastics of microplastics in seawater, and are thus central to source attribution (Borah et al., 2023; Hurley et al., 2025). The researchers will rinse the collected plastic with Hyclin plus, tap water, and Milli-Q distilled water. The plastics are then patted dry with a clean cloth. After sorting into categories based on the three most prevalent plastic items in the study area, researchers will carefully cut the plastics into pieces no larger than 5 mm to retain their microplastic properties. The researchers will use a Vernier caliper to cut the collected plastics to ensure all plastic samples are the same size. The goal is to create semi-square or rectangular

pieces, depending on the material limitations (Villegas-Camacho et al., 2024). Next, plastic sample classification will be based on nine common plastic polymers: polyethylene terephthalate (PET), high-density polyethylene (HDPE), polyvinyl chloride (PVC), low-density polyethylene (LDPE), polypropylene (PP), polystyrene (PS), polyamide (PA), multilayers, and others (Pacilan & Bacosa, 2022).

In this study, Barangay Camagong seawater will be selected for three sampling sites to allow researchers to collect macroplastics that may contribute to microplastic pollution through environmental degradation. These sites will be strategically selected based on proximity to potential pollution sources, including residential, non-residential, and mangrove areas. The collected macroplastics will be categorized by type, size, and the three most common items to help trace their possible origins and confirm their potential to fragment into microplastics over time.

2.5 Research Ethical Standards

A. Intellectual Property Rights (IPR)

Intellectual property rights will be upheld in this study, ensuring that all materials to create, including tools for data gathering, techniques, and analytical frameworks, will be protected under Philippine copyright law. The study will not involve the development of new technologies, products, or inventions that require patent protection. However, all original outputs, such as data collection tools, methods, graphs, and reports, will be protected under copyright as the intellectual property of the researchers and Saint Michael College of Caraga. Any future use of the study materials will include proper acknowledgment of the authors.

B. Informed Consent

Before data collection, the researchers will seek permission from the Barangay Captain of Camagong and the Municipal Environment and Natural Resources Office (MENRO) of Nasipit. The informed consent for the boat driver during the boat transect and seek permission to use their time, which the researchers will compensate them for the time and effort to participate in the collection of plastics. The researchers will inform the residents of the study's purpose and the activities to be conducted along the shoreline. The researchers will only include areas with local authority consent in the sampling. No human participants will be directly tested or subjected to experimentation in this research.

C. Data Privacy and Confidentiality

No personal information from residents will be included in this study. All data related to sampling locations, plastic types, and collected quantities will be stored securely and used solely for academic purposes. Only authorized members of the research team will have access to the data.

D. Voluntary Participation

Participation and cooperation from local authorities and community members will be voluntary. They may withdraw their consent at any time without penalty. The researchers will not provide monetary incentives or gifts to ensure impartiality and fairness.

E. Minimization of Harm

The researchers will collect macroplastics responsibly to prevent harm to marine life and coastal ecosystems. Researchers will wear Personal Protective Equipment (PPE), such as gloves, masks, surgical gowns, goggles, face shields, and boots, to maintain safety and prevent

contamination. After analysis, all collected plastic waste will be properly segregated and disposed of in accordance with the Municipal Solid Waste Management Guidelines.

F. Beneficence

The research will aim to contribute positively to the coastal community of Barangay Camagong, Nasipit, Agusan del Norte, by providing essential information about the presence, types, and sources of macroplastic pollution along its shoreline. The results of this study will have potential long-term ecological and public health benefits for locals by supporting regional environmental initiatives and providing guidance for improving coastal waste management practices. The Department of Environment and Natural Resources Community Environment and Natural Resources Office (DENR-CENRO) Nasipit and the Local Government Unit (LGU) of Nasipit may utilize the findings to enhance solid waste management systems and reduce plastic leakage into marine environments.

G. Justice and Fair Participant Selection

Site selection will be based solely on environmental relevance, not socioeconomic or demographic characteristics. The researchers will ensure fair and unbiased treatment of all individuals involved during community coordination, upholding inclusivity and equal representation.

H. Data Integrity and Accuracy

Standardized scientific procedures for collecting, sorting, and measuring macroplastics will be followed to ensure reliability and validity. Data entries will be double-checked and verified for consistency. All possible biases or limitations encountered during fieldwork will be recorded and discussed transparently in the report.

I. Transparency in Reporting

The results of the study will be presented truthfully and openly. Researchers will report both positive and negative outcomes, and remain committed to presenting results honestly and with integrity, including the sponsorships and donations being offered to the researchers as to help with the expenses during the testing of samples. The researchers will cite all references and data sources correctly to maintain transparency and protect intellectual property.

J. Use of Patented or Copyrighted Materials

The researchers will not use any patented or copyrighted materials without permission. Any literature, figures, or tools borrowed from other works will be cited appropriately. The researchers will respect all third parties' intellectual property rights throughout the study.

K. Ethical Considerations for Human and Environmental Interaction

No human or animal subjects will be used in experimental procedures. Environmental ethics will be observed by ensuring that no harm is caused to the natural surroundings during sampling. The research team will follow local and institutional guidelines for responsible ecological research practices.

L. Responsible Use of AI and Related Technologies

Artificial Intelligence (AI) tools will be used only for grammar checking, formatting, and organization of written output. The researchers will refrain from using AI to generate research data or manipulate results. The researchers will ensure that the study's integrity, originality, and authorship are maintained at all times.

M. Ethical Approval

The research proposal will be submitted to the Research Ethics Committee of Saint Michael College of Caraga for review and approval before fieldwork. The study will comply with institutional and national ethical standards governing environmental and scientific research.

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